

Econometrics: What's It All About?

Using sample data on observable variables to learn about economic relationships, the functional relationships among economic variables.

Econometrics consists mainly of:

- **estimating** economic relationships from sample data
- **testing hypotheses** about how economic variables are related
 - the **existence of relationships** between variables
 - the **direction of the relationships** between one economic variable -- the dependent or outcome variable -- and its hypothesized observable determinants
 - the **magnitude of the relationships** between a dependent variable and the independent variables that are thought to determine it.

Sample data consist of **observations on randomly selected members of populations** of economic agents (individual persons, households or families, firms) or other units of observation (industries, provinces or states, countries).

Example 1

We wish to investigate empirically the **determinants** of households' food expenditures, in particular the relationship between households' food expenditures and households' incomes.

Sample data consist of **a random sample of 38 households** from the population of all households. For each household in the random sample, we have observations on three observable variables:

foodexp = annual food expenditure of household, thousands of dollars per year
income = annual income of household, thousands of dollars per year
hhsize = household size, number of persons in household

```
. list foodexp income hhsize
```

	foodexp	income	hhsize
1.	15.998	62.476	1
2.	16.652	82.304	5
3.	21.741	74.679	3
4.	7.431	39.151	3
5.	10.481	64.724	5
6.	13.548	36.786	3
7.	23.256	83.052	4
8.	17.976	86.935	1
9.	14.161	88.233	2
10.	8.825	38.695	2
11.	14.184	73.831	7
12.	19.604	77.122	3
13.	13.728	45.519	2
14.	21.141	82.251	2
15.	17.446	59.862	3
16.	9.629	26.563	3
17.	14.005	61.818	2
18.	9.16	29.682	1
19.	18.831	50.825	5
20.	7.641	71.062	4
21.	13.882	41.99	4
22.	9.67	37.324	3
23.	21.604	86.352	5
24.	10.866	45.506	2
25.	28.98	69.929	6
26.	10.882	61.041	2
27.	18.561	82.469	1
28.	11.629	44.208	2
29.	18.067	49.467	5
30.	14.539	25.905	5
31.	19.192	79.178	5
32.	25.918	75.811	3
33.	28.833	82.718	6
34.	15.869	48.311	4
35.	14.91	42.494	5
36.	9.55	40.573	4
37.	23.066	44.872	6
38.	14.751	27.167	7

```
. describe
```

Contains data from foodexp.dta

obs:	38	
vars:	3	7 Sep 2020 23:30
size:	608 (99.9% of memory free)	

1. foodexp	float	%9.0g	food expenditure, thousands \$ per yr
2. income	float	%9.0g	household income, thousands \$ per yr
3. hhsize	float	%9.0g	household size, persons per hh

Sorted by:

Note: dataset has changed since last saved

```
. summarize
```

Variable	Obs	Mean	Std. Dev.	Min	Max
foodexp	38	15.95282	5.624341	7.431	28.98
income	38	58.44434	19.93156	25.905	88.233
hhsz	38	3.578947	1.702646	1	7

Question: What relationship generated these sample data? What is the data generating process?

Answer: We postulate that **each population value of foodexp**, denoted as **foodexp_i**, is generated by a relationship of the form:

$$\text{foodexp}_i = f(\text{income}_i, \text{hhsz}_i) + u_i \quad \Leftarrow \text{the population regression equation}$$

where

food exp_i = the **dependent or outcome variable** we are trying to explain
= the annual food expenditure of household i (thousands of \$ per year)

income_i = an **independent or explanatory variable** that we think might explain the dependent variable food exp_i
= the annual income of household i (thousands of \$ per year)

hhsz_i = a second **independent or explanatory variable** that we think might explain the dependent variable food exp_i
= household size, measured by the number of persons in the household

$f(\text{income}_i, \text{hhsz}_i)$ = a **population regression function** representing the systematic relationship of food exp_i to the independent or explanatory variables income_i and hhsz_i;

u_i = an **unobservable random error term** representing **all unknown and unmeasured variables** that determine the individual population values of food exp_i

Question: What mathematical form does the population regression function $f(\text{income}_i, \text{hhsz}_i)$ take?

Answer: We hypothesize that the **population regression function** -- or **PRF** -- is a linear function:

$$f(\text{income}_i, \text{hhsz}_i) = \beta_0 + \beta_1 \text{income}_i + \beta_2 \text{hhsz}_i$$

Implication: The **population regression equation** -- the **PRE** -- is therefore

$$\begin{aligned} \text{food exp}_i &= f(\text{income}_i, \text{hhsz}_i) + u_i \\ &= \beta_0 + \beta_1 \text{income}_i + \beta_2 \text{hhsz}_i + u_i \end{aligned}$$

- **Observable Variables:**

$\text{food exp}_i \equiv$ the value of the dependent variable food exp for the i -th household

$\text{income}_i \equiv$ the value of the independent variable income for the i -th household

$\text{hhsz}_i \equiv$ the value of the independent variable hhsz for the i -th household

- **Unobservable Variable:**

$u_i \equiv$ the value of the random error term for the i -th household in the population

- **Unknown Parameters:** the **regression coefficients** β_0 , β_1 and β_2

$\beta_0 =$ the *intercept coefficient*

$\beta_1 =$ the *slope coefficient on income*

$\beta_2 =$ the *slope coefficient on hhsz*

The **population values** of the regression coefficients β_0 , β_1 and β_2 are *unknown*.

Example 2

We wish to investigate empirically the determinants of paid workers' wage rates. In particular, we want to investigate whether male and female workers with the same characteristics on average earn the same wage rate.

Sample data consist of **a random sample of 526 paid workers** from the 1976 US population of all paid workers in the employed labour force.

For each paid worker in the random sample, we have observations on six observable variables:

wage = average hourly earnings of paid worker, dollars per hour
ed = years of education completed by paid worker, years
exp = years of potential work experience of paid worker, years
ten = tenure, or years with current employer, of paid worker, years
female = 1 if paid worker is female, = 0 otherwise
married = 1 if paid worker is married, = 0 otherwise

. describe

Contains data from wage1.dta

obs: 526
vars: 6 16 Apr 2020 16:18
size: 94,680 (90.7% of memory free)

1. wage	float	%9.0g	average hourly earnings, \$/hour
2. ed	float	%9.0g	years of education
3. exp	float	%9.0g	years of potential work experience
4. ten	float	%9.0g	tenure = years with current employer
5. female	float	%9.0g	=1 if female, =0 otherwise
6. married	float	%9.0g	=1 if married, =0 otherwise

```
. list wage ed exp ten female married
```

	wage	ed	exp	ten	female	married
1.	21.86	12	24	16	0	1
2.	5.5	12	18	3	0	0
3.	3.75	2	39	13	0	1
4.	10	12	31	2	0	1
5.	3.5	13	1	0	0	0
6.	6.67	12	35	10	0	0
7.	3.88	12	12	3	0	1
8.	5.91	12	14	6	0	1
9.	5.9	12	14	7	0	1
10.	10	17	5	3	0	1
11.	4.55	16	34	2	0	1
12.	10	8	9	0	0	1
13.	6	13	8	0	0	1
14.	5	9	31	9	0	1
15.	4.5	12	13	0	0	1
16.	5.43	14	10	3	0	1
17.	2.83	10	1	0	0	0
18.	6.8	12	14	10	0	1
19.	6.76	12	19	3	0	1
20.	4.51	12	5	2	0	0

```
(output omitted)
```

507.	6.15	12	35	12	1	0
508.	11.1	15	1	4	1	0
509.	3.35	15	3	1	1	1
510.	5	12	7	3	1	0
511.	3.35	7	35	0	1	0
512.	6.25	12	13	0	1	1
513.	3.06	12	14	10	1	0
514.	5.9	12	9	7	1	1
515.	8.1	12	38	3	1	1
516.	14.58	18	13	7	1	0
517.	9.42	14	23	0	1	1
518.	9.68	13	16	16	1	0
519.	8.6	16	3	2	1	0
520.	3	12	38	0	1	1
521.	3.33	12	45	4	1	1
522.	4	12	22	11	1	1
523.	2.75	13	1	2	1	0
524.	3	16	19	10	1	1
525.	2.9	8	49	6	1	0
526.	3.18	12	5	0	1	1

```
. summarize wage ed exp ten female married
```

Variable	Obs	Mean	Std. Dev.	Min	Max
wage	526	5.896103	3.693086	.53	24.98
ed	526	12.56274	2.769022	0	18
exp	526	17.01711	13.57216	1	51
ten	526	5.104563	7.224462	0	44
female	526	.4790875	.500038	0	1
married	526	.608365	.4885804	0	1

Question: What relationship generated these sample data? What is the data generating process?

Answer: We postulate that **each population value of wage**, denoted as **wage_i**, is generated by a **population regression equation** of the form:

$$\text{wage}_i = f(\text{ed}_i, \text{exp}_i, \text{ten}_i, \text{female}_i, \text{married}_i) + u_i$$

where:

wage_i = the **dependent or outcome variable** we are trying to explain

= the average hourly earnings of paid worker i (dollars per hour)

ed_i = one **independent or explanatory variable** that we think might explain the dependent variable wage_i

= the years of education completed by paid worker i (years)

exp_i = a second **independent or explanatory variable** that might explain wage_i

= the potential work experience accumulated by paid worker i (years)

ten_i = a third **independent or explanatory variable** that might explain wage_i

= tenure, years with current employer, of paid worker i (years)

female_i = a fourth **independent or explanatory variable** that might affect wage_i

= 1 if paid worker i is female, = 0 otherwise

married_i = a fifth **independent or explanatory variable** that we think might explain the dependent variable wage_i

= 1 if paid worker i is married, = 0 otherwise

f(ed_i, exp_i, ten_i, female_i, married_i)

= a **population regression function** representing the systematic relationship of wage_i to the independent variables ed_i, exp_i, ten_i, female_i and married_i

u_i = an **unobservable random error term** representing **all unknown variables and unmeasured** that determine the individual population values of wage_i

Question: What mathematical form does the population regression function, or PRF, $f(\text{ed}_i, \dots, \text{married}_i)$ take?

Answer: We hypothesize that the **population regression function** -- or **PRF** -- is a **linear function**.

$$f(\text{ed}_i, \dots, \text{married}_i) = \beta_0 + \beta_1 \text{ed}_i + \beta_2 \text{exp}_i + \beta_3 \text{ten}_i + \beta_4 \text{female}_i + \beta_5 \text{married}_i$$

Implication: The **population regression equation** -- the **PRE** -- is therefore

$$\begin{aligned} \text{wage}_i &= f(\text{ed}_i, \text{exp}_i, \text{ten}_i, \text{female}_i, \text{married}_i) + u_i \\ &= \beta_0 + \beta_1 \text{ed}_i + \beta_2 \text{exp}_i + \beta_3 \text{ten}_i + \beta_4 \text{female}_i + \beta_5 \text{married}_i + u_i \end{aligned}$$

- **Observable Variables:**

$\text{wage}_i \equiv$ the value of the dependent variable wage for the i -th employee

$\text{ed}_i \equiv$ the value of the independent variable ed for the i -th employee

$\text{exp}_i \equiv$ the value of the independent variable exp for the i -th employee

$\text{ten}_i \equiv$ the value of the independent variable ten for the i -th employee

$\text{female}_i \equiv$ the value of the independent variable female for the i -th employee

$\text{married}_i \equiv$ the value of the independent variable married for the i -th employee

- **Unobservable Variable:**

$u_i \equiv$ the value of the random error term for the i -th paid worker in the population

- **Unknown Parameters:** the regression coefficients $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$ and β_5

$\beta_0 =$ the *intercept* coefficient

$\beta_1 =$ the *slope* coefficient on ed_i

$\beta_2 =$ the *slope* coefficient on exp_i

$\beta_3 =$ the *slope* coefficient on ten_i

$\beta_4 =$ the *slope* coefficient on female_i

$\beta_5 =$ the *slope* coefficient on married_i

Our task: To learn how to compute from sample data reliable estimates of the regression coefficients $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$ and β_5 .

The Four Elements of Econometrics

Data

Collecting and coding the sample data, the raw material of econometrics.

Most economic data is **observational, or non-experimental, data** (as distinct from **experimental data** generated under controlled experimental conditions).

Specification

Specification of the *econometric model* that we think (hope) generated the sample data -- that is, specification of the **data generating process** (or DGP).

An ***econometric model*** consists of **two** components:

1. An ***economic model***: specifies the **dependent or outcome variable** to be explained and the **independent or explanatory variables** that we think are related to the dependent variable of interest.
 - Often suggested or derived from economic theory.
 - Sometimes obtained from informal intuition and observation.
2. A ***statistical model***: specifies the statistical elements of the relationship under investigation, in particular the **statistical properties of the random variables** in the relationship.

Estimation

Consists of **using the assembled sample data** on the **observable variables** in the model **to compute estimates** of the **numerical values** of all the **unknown parameters** in the model.

Inference

Consists of **using the parameter estimates** computed from sample data **to test hypotheses** about the **numerical values** of the **unknown population parameters** that describe the behaviour of the population from which the sample was selected.

Scientific Method

The collection of principles and processes necessary for scientific investigation, including:

1. rules for concept formation
2. rules for conducting observations and experiments
3. rules for validating hypotheses by observations or experiments

Econometrics is that branch of economics -- the dismal science -- which is concerned with items 2 and 3 in the above list.



Recap

We have considered **two examples** of what are generically called ***linear regression equations*** or ***linear regression models***.

Example 1 -- a linear regression model for household food expenditure:

$$\text{food exp}_i = \beta_0 + \beta_1 \text{income}_i + \beta_2 \text{hhsz}_i + u_i$$

Example 2 -- a linear regression model for paid workers' wage rates:

$$\text{wage}_i = \beta_0 + \beta_1 \text{ed}_i + \beta_2 \text{exp}_i + \beta_3 \text{ten}_i + \beta_4 \text{female}_i + \beta_5 \text{married}_i + u_i$$

Regression analysis has two fundamental tasks:

1. **Estimation**: computing from *sample data* reliable *estimates* of the *numerical values of the regression coefficients* β_j ($j = 0, 1, \dots, K$), and hence of the population regression function.
2. **Inference**: using *sample estimates* of the regression coefficients β_j ($j = 0, 1, \dots, K$) to test hypotheses about the *population values* of the **unknown regression coefficients** -- i.e., to *infer from sample estimates* the **true population values of the regression coefficients** *within specified margins of statistical error*.